

ESP32 BLE DUAL JOYSTICK GAME CONTROLLER

1. Project Overview

This is a personal DIY electronics project to build a wireless, PS5-style Bluetooth game controller using an ESP32 microcontroller. The goal was to make a fully custom gamepad from scratch — using basic components like joystick modules and push buttons — that connects wirelessly to a PC, phone, or browser-based game via Bluetooth, just like a real gaming controller.

2. Motivation

Wanted to understand how commercial game controllers work internally, and build a working, playable version myself instead of buying one — combining basic electronics with embedded programming and wireless communication.

3. Components Used

Component	Quantity
ESP32 DevKit V1	1
Analog Joystick Module	2
Push Buttons (4-pin tactile)	9
Breadboard	1
Jumper Wires	As needed
USB Cable (power/programming)	1

4. How It Works

- Each joystick outputs analog voltage (X and Y) read through the ESP32's built-in ADC.
- Each push button is read as a simple digital HIGH/LOW signal using internal pull-up resistors — no external resistors needed.
- The ESP32 packages all this input data into a Bluetooth Low Energy (BLE) HID gamepad report, using the **ESP32 BLE Gamepad** library.
- Once paired, the ESP32 shows up on any Bluetooth-enabled device exactly like a standard wireless game controller — no drivers needed.

5. Final Pin Configuration

Left Joystick: X → GPIO34, Y → GPIO35, Click (L3) → GPIO32 **Right Joystick:** X → GPIO33, Y → GPIO36, Click (R3) → GPIO25

9 Push Buttons: GPIO23, GPIO22, GPIO21, GPIO19, GPIO18, GPIO27, GPIO14, GPIO12, GPIO13

6. Software Used

- Arduino IDE
- ESP32 Board Package
- ESP32 BLE Gamepad library (by lemmingDev)
- NimBLE-Arduino (dependency)

7. Problems Faced & How They Were Solved

This section is the most useful part of building anything — here's what actually went wrong and how it got fixed:

- **4-pin push buttons were confusing at first** — turned out the 4 legs are just 2 internally-connected pairs, not 4 separate connections. Solved by using a multimeter/continuity check and wiring one leg from each side (diagonal pins).
- **Windows kept detecting the controller as "2 axis 7 button gamepad"** even after updating the code to 4 axes/9 buttons — this was due to Windows caching the old Bluetooth HID descriptor. Fixed by fully removing the Bluetooth pairing, uninstalling the HID driver from Device Manager, restarting Windows, and re-pairing fresh.
- **4 specific buttons (right-side group) weren't responding** — traced to a wiring issue in that group (likely a loose GND connection on the breadboard), isolated using a simple standalone digitalRead() test sketch before touching the full BLE code again.
- **Confusion about which side of the ESP32 pins were physically located on the board** — resolved by focusing on matching GPIO numbers exactly rather than relying on "left/right" labels, since silkscreen layout varies slightly between board manufacturers.

8. Testing Method

- Used gamepad-tester.com to verify Bluetooth HID input recognition (all axes and button presses) directly in-browser before testing in actual games.
- Used Windows' built-in Game Controller settings (joy.cpl) to double check button/axis count matched the code.
- Verified compatibility with Need for speed most wanted 2012 controller-supported game section as a real-world test.

9. Current Status

Fully working prototype on breadboard — all 2 joysticks and 9 buttons responding correctly over Bluetooth, verified working in browser-based games with controller support.

10. Planned Future Improvements

- Move from breadboard to a proper enclosure (small plastic project box)
- Add battery power (Li-ion + charging module) for a fully wireless, portable build
- Add L2/R2 trigger buttons
- Add basic vibration/haptic feedback
- Add an OLED screen for battery status/connection status
- 3D print a custom shell once the final layout is confirmed